title: "External Audit of OpenAI's Quasi-Riemann Hypothesis Manuscript" subtitle: "Master audit, loss table and residual-risk notes" date: "October 2026" geometry: margin=2.2cm fontsize: 10pt

abstract: "This is an external audit of the existing manuscript, not an independent re-proof of the analytic estimates. Primary objects: OpenAI, The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane Re(s)>7/8 (preprint, 30 Sep 2026) and its Lean formalization, github.com/openai/math."

Independent Multi-Agent Audit of the OpenAI Quasi-Riemann Hypothesis Manuscript

Date: 2026-10-13
Method: Layer-by-layer reconstruction and stress-test using SuperGrok Heavy multi-agent analysis of the public TeX source and Lean formalization.

Claims (precise scope)

1. The logical architecture of the proof is coherent at every examined layer.
2. The paper's own exponent ledgers for the modest positive margins (Stage 1 1/4800, Stage 2 endpoint 49/440640) close with a positive remainder under the losses they explicitly charge.
3. The Lean formalization of the final theorems aligns with the claimed statements; the final theorems are machine-checked, and an independent third-party kernel replay has been reported (see Part 9).
4. No structural gap or disproof has been identified.

What this is not: A complete independent proof of the analytic estimates. The residual risk is quantitative and localized to the possibility of an unlisted loss in an intermediate lemma.

Architecture (Parts 1–9)

Part	Layer	Finding
1	Continuation criterion	Dual representations produce contradiction if zero lies too far right. Structural soundness confirmed.
2	Cubic-theta reflection	Exact under stated conditions; retains zero-on-nonunit restrictions.
3	Poisson & principal signal	Principal $1/L$ Mellin signal cleanly isolated.

Part	Layer	Finding
4	Large sieve & row energy	Specialized sextic sieve + envelope $R(\delta)$ supply the positive margins.
5	Zero detector & moments	Saturation forces inverse witness dominant; feeds the envelope.
6	Stage 1 assembly (11/12)	Balanced scales, $C(s) = s - 2/3$, margin $1/4800$ under stated accounting.
7	Stage 2 refinement (7/8)	Compensation + asymmetric scales, endpoint positive.
8	Uniformity & transfer	Margins independent of target; quadratic factorization transfers to Dirichlet L-functions.
9	Lean alignment	Final theorems formalized and kernel-checked; third-party replay reported.

Specialist quantitative audit

Stage 1 high-side saving

- Reserved: $m_0 = 1/1200$, $\epsilon_* = 10^{-6}$, $m = 1/2400$.
- Central losses: $46\epsilon_* < 1/4800$.
- Principal and small-row losses each $< m_0/4$.
- After fixed scalar and late-height closure: claimed remainder $1/4800$.
- Status: table closes with positive remainder under charged losses.

Stage 2 endpoint certificate

- Algebraic identity yields $-E_* \geq 49/440640 > 1/10000$.
- Additional losses charged to reserved fraction of high-side margin or arbitrarily small ϵ_{total}.
- Status: remainder positive for every $\Delta = \beta_* - 7/8 > 0$.

Computational confirmation

- Row envelope $R(\delta) = \min\{1, \max(1 - \delta/2, 4/3 - \delta)\}$ verified; strict saving for $\delta > 1/3$.
- Endpoint constant $49/440640 \approx 0.0001112 > 1/10000$ confirmed.

Lean skeleton of the hinge

The continuation criterion has been isolated as a pure logical implication: dual bounds with uniform positive powers imply holomorphic continuation of $1/L$ past the supposed rightmost zero. The concrete margins appear as explicit constants. The analytic estimates that discharge the implication remain in the prose.

Residual risk

The margins are small. The audit confirms they survive every loss the paper explicitly charges and that the computational constants match. The residual risk is the existence of an unlisted intermediate loss larger than the reserved budget. Full certification requires specialist verification of every lemma that feeds the ledger, or formalization of those steps.

Conclusion

The manuscript survives structural scrutiny, Lean alignment, and specialist audit of its own exponent accounting. No disproof has been found. The quantitative hinge is localized and, under the stated accounting, closes with a positive remainder. The paper's own accounting is internally consistent under its methods; the result has not been independently re-proved from first principles.

Artifacts

- [01_continuation_criterion.md] (01_continuation_criterion.md) – [09_lean_alignment.md] (09_lean_alignment.md)
- [10_specialist_margins.md] (10_specialist_margins.md)
- [11_loss_table.md] (11_loss_table.md)
- [17_computational_check.md] (17_computational_check.md)
- This document

Specialist Loss Table

Stage 1 (11/12) and Stage 2 (7/8) endpoint certificates

Purpose: Self-contained extraction of every charged loss that feeds the final positive margins, with the reserved budgets they are charged against.

Stage 1 — High-side saving $1/4800$

Source: Proposition on balanced-high estimate and Lemma on late-height-closure.

Reserved parameters

Parameter	Value	Role
m_0	$1/1200$	Minimum error-free saving on every major contribution
ϵ_*	10^{-6}	Unit loss charged to each listed class
m	$1/2400$	Saving used after preliminary losses
ω	$\Delta_1/2$	Low-bound room

Charged losses (explicit count)

Loss class	Coefficient	Charged amount	Reserved budget
Central (row count, numerator, local correction, reciprocal, multiplicity, physical dyads)	$13\epsilon_* + 2(15\epsilon_*) + 3\epsilon_*$	$46\epsilon_*$	$< m_0/4 = 1/4800$
Principal residues (two terms)	$\le (3/2+1)\epsilon_*$ and $2\epsilon_*$	each $< m_0/4$	$m_0/4$
Small-row	$\le 2\epsilon_*$	$< m_0/4$	$m_0/4$
Large-row	requested ϵ_*	absorbed	reserved
Height factor $(1+T_1)^{A_\eta}$	late choice of τ	absorbed into half remaining margin	late-height closure
Contour tails $O(Z^{B_\eta} T_1^{-N})$	increase N after target	smaller than any positive power	external order

Accounting

- After all listed losses and division by the fixed scalar c_S, at least $3m_0/4 = 1/1600$ remains before the height factor.
- Using $m = 1/2400$ leaves an unused $1/4800$.
- Late-height closure absorbs the height factor into half the remainder.
- Final claimed high-side saving: $1/4800$.

Status: Every listed loss is strictly smaller than its reserved fraction under the paper's accounting. The table closes with a positive remainder.

Stage 2 — Endpoint certificate 49/440640

Source: Lemma on balanced-endpoint (or equivalent high-bin endpoint calculation).

Certificate derivation (extracted) The high exponent at the endpoint satisfies an identity of the form

$$10368\,\mathcal{J}\,(-E_*) = (\text{non-negative quadratic form in } \delta, y).$$

The leading constant term is 49. Bounding the remaining factors ($\mathcal{J} \le 5/2$, $v \le 17(3+5y)$) produces

$$-E_* \ge \frac{49}{440640} > \frac{1}{10000}.$$

The inequality holds uniformly for $1/50 < \delta \le 5/6$.

Charged losses against this certificate

Loss class	Treatment	Budget
Amplitude bin width ϑ, witness subdivisions	absorbed into $O(\epsilon)$	reserved fraction of high margin
Capacity reduction (ν_0)	charged $2q\nu_0 + \delta \max w_i$	$\Delta/4$ or equivalent
Height factor	late τ as in Stage 1	half high margin
Contour tails	external order N	smaller than any positive power
Compensation and asymmetric dyadics	absorbed into same ϵ_{total}	target-independent coefficient

Accounting

- The constant term alone already exceeds 1/10000.
- All additional losses are charged to a reserved fraction of the high-side margin (proportional to $\Delta = \beta_* - 7/8$) or to an arbitrarily small ϵ_{total}.
- The certificate remains positive for every $\Delta > 0$.

Status: The endpoint inequality is negative by a fixed positive amount plus a term proportional to Δ. Under the paper's accounting the remainder stays positive.

Overall specialist conclusion

Both certificates close with a positive remainder under the losses the paper explicitly charges. The Stage 1 table accounts for every listed ϵ_* class and leaves 1/4800. The Stage 2 certificate supplies a fixed positive constant larger than 1/10000 that survives the same style of reserved-budget accounting. No listed

loss exceeds its budget in the extracted text. The quantitative hinge survives this level of extraction.

Remaining limitation: The table uses the paper's own charged amounts. Full certification still requires confirming that no intermediate lemma contains an additional unlisted loss of larger size. Within the extracted accounting, the margins hold.

Residual-Risk Notes

Editorial index compiled for this repository. Every statement below is taken from the audit files linked; nothing here adds a new finding.

Where the risk sits

The margins are small: a Stage 1 high-side saving of $1/4800$ and a Stage 2 endpoint constant $49/440640 \approx 0.0001112 > 1/10000$. The audit confirms that these survive every loss the paper **explicitly** charges and that the computational constants match ([00] (00_master_audit.md), [11] (11_loss_table.md), [17] (17_computational_check.md)). The residual risk is the existence of an **unlisted** intermediate loss larger than the reserved budget. Confirming that none exists requires a specialist reading of every lemma that feeds the ledger, or formalization of those steps.

Item-by-item

Item	What could fail	Audit file
Stage 1 central loss $46\epsilon_* < m_0/4$	Additional loss classes in the full central-contribution equation not included in the 13 or the two 15s	[12] (12_stage1_central_loss.md)
Late-height closure	An estimate producing A_η that itself depends on a height not yet closed	[13] (13_late_height_closure.md)
Specialized sextic large sieve $(KD)^{2/3}$	An extra factor in the full write-up not absorbed into ϵ; an error in the mod-6 recursion would degrade $R(\delta)$	[14] (14_sextic_large_sieve.md), [04] (04_large_sieve_row_energy.md)
Zero-detector saturation	Product lower bound weaker than claimed	[15] (15_detector_saturation.md), [05] (05_zero_detector_moments.md)
Stage 2 compensation identities	An uncancelled, target-dependent main term in an intermediate expansion	[16] (16_stage2_compensation.md), [07] (07_stage2_refinement_7_8.md)

Item	What could fail	Audit file
Uniformity in the conductor	A slowly growing conductor factor hidden inside an ϵ	[08] (08_uniformity_transfer.md), [01] (01_continuation_criterion.md)
Lean translation fidelity	Lean certifies the final theorems, not every intermediate prose estimate	[09] (09_lean_alignment.md)

What would settle it

A short specialist note either confirming that the listed losses are fully accounted for under the paper's methods, or naming a specific lemma and loss that exceeds the reserved budget. See [18] (18_expert_review_request.md).